

The Re-prompt Tax: Measuring Hidden Interaction Costs for Multilingual and Code-Switched LLM Users

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Abstract

Multilingual users often interact with LLMs through code-switched prompts, mixed-language editing requests, and constraints about script, register, locale, and literal text preservation. Standard one-turn evaluations miss a common usability failure: the model returns a fluent, plausible answer that violates the user’s interaction contract, forcing the user to spend extra turns repairing the response. We introduce *Re-prompt Tax*, a metric family that measures the turns, tokens, and repair success required to recover from these failures. We construct a 120-item stress pilot across Spanish-English, Hindi-English, and Arabic-English interactions and evaluate three GPT-4.1-family API models under a baseline prompt and a Global Interaction Contract prompt. Baseline first-turn global alignment ranges from 67.5% to 76.7%; the mitigation improves alignment to 76.7–93.3% and reduces mean repair turns. A 10% stratified blinded LLM-judge audit agrees with the automatic scorer on 98.6% of sampled pass/fail labels.

1 Introduction

The GenAI4World workshop asks how evaluations should capture global tasks, contexts, vulnerabilities, and usability beyond standard accuracy metrics (GenAI4World Organizers, 2026). We study one such gap: a model can return a semantically plausible answer while violating the user’s multilingual interaction contract. The failure may look minor in a one-turn benchmark, but it imposes an extra interaction burden on the user.

For example, a user may ask in Spanish:

Me ayudas a mejorar este texto para que suene natural? "I can't join the call today, but I can share my update by email."

A globally aligned assistant should infer that the quoted English sentence is the object of editing and should remain English unless translation is requested. In our pilot, models sometimes answer with Spanish preambles or full Spanish translations. The answer may be fluent and helpful-looking, but the user still has to re-prompt: "Please keep the rewritten text in English."

We call this burden the *Re-prompt Tax*. Our contributions are:

- We define metrics for hidden multilingual repair burden: first-turn global alignment, re-prompt turn tax, repair success, unresolved rate, and token tax.
- We build a small, privacy-safe stress pilot spanning language, script, literal-preservation, register, and locale constraints.
- We evaluate three GPT-4.1-family API models (OpenAI, 2025) under baseline and mitigation conditions, then validate a stratified sample with a blinded LLM judge.

As a lightweight motivation check, we also scan the first 20,000 streamed WildChat conversations (Zhao et al., 2024) using predefined repair cues. Among 10,681 multi-turn conversations, 172 contain at least one follow-up cue related to generic repair, preservation, output language, unwanted translation, script, or register/locale. We use this only to motivate the taxonomy, not as a representative prevalence estimate.

2 Re-prompt Tax

Let item i contain a user prompt, expected output language and script, spans that must be preserved exactly, and any register or locale requirements. We define **First-Turn Global Alignment**:

$$\text{FTGA}(i, m) = 1$$

if model m satisfies all expected language, script, preservation, task, register, and locale constraints on the first response, and 0 otherwise.

If the first response fails, we append up to two standardized repair prompts. **Re-prompt Turn Tax** (RTT) is the minimum number of repair prompts needed before success:

$$\text{RTT}(i, m) \in \{0, 1, 2, 3\},$$

where 3 denotes unresolved after two repairs. We also report Repair@1, Repair@2, unresolved rate, and token tax: total tokens consumed until success, divided by first-attempt tokens.

This framing is related to output-language alignment in code-switched interactions (Oh et al., 2026), but differs in two ways. First, we measure repair trajectories rather than only first-turn outcomes. Second, the contract includes not only output language, but also script, literal span preservation, register, and locale-sensitive data.

3 Benchmark and protocol

Items. We created REPROMPTTAX-STRESS-V0.2, a 120-item pilot with three language pairs: Spanish-English, Hindi-English, and Arabic-English. Each pair has ten items in four task families: implicit editing preservation, output-language inference, quote preservation, and script/register/locale constraints. Items are synthetic and privacy-safe, but were designed to reflect repair patterns that appear plausible in real multilingual chat use.

Models and prompts. We evaluate gpt-4.1-nano, gpt-4.1-mini, and gpt-4.1 with temperature 0. Each model is run under two conditions. The baseline system prompt is “You are a helpful assistant.” The mitigation condition uses a Global Interaction Contract: infer the language of the instruction and content, preserve content language during editing unless translation is requested, preserve quoted spans and literal data, obey explicit script/register/locale constraints, and avoid unnecessary explanation.

Scoring and audit. Automatic scoring combines deterministic checks for exact span preservation and script with rule-based language and task checks. We then ran a blinded gpt-4.1 judge audit on 72 first-turn responses, sampling three responses from each model-condition-family stratum. The judge saw only the user prompt, expected contract, and assistant response, not model identity or condition. Experiments were run on June 28, 2026.

4 Results

Table 1 shows measurable baseline re-prompt tax for all three models. The mitigation improves FTGA for every model, reduces mean RTT and mean token tax, and brings the strongest model to 93.3% FTGA, though residual exact-preservation failures remain. Paired bootstrap estimates over the same 120 items show FTGA gains of +9.2, +3.3, and +16.7 points for nano, mini, and GPT-4.1; mean token tax falls by $0.35\times$, $0.16\times$, and $0.30\times$. Single-model prompt diagnostics on nano show that a generic helpfulness prompt improves FTGA from 67.5% to 75.0%, while the full contract reaches 76.7%. A narrower content-preservation prompt reaches 80.0% FTGA, 0.27 mean RTT, and the lowest token tax, so we treat the full contract as the pre-registered three-model intervention, not as the best possible prompt. Among first-turn failures, 31/96 baseline first-turn failures still pass the task component; the stricter task-plus-preservation slice falls from 11 to 1 under the contract, isolating repair burden that is not simple task failure.

Model	Condition	FTGA	Mean RTT	Tok. tax	Unres.	R@1	R@2
GPT-4.1 nano	Baseline	67.5	0.47	1.69	5.8	74.4	82.1
GPT-4.1 nano	Contract	76.7	0.33	1.34	4.2	78.6	82.1
GPT-4.1 mini	Baseline	75.8	0.28	1.43	0.8	89.7	96.6
GPT-4.1 mini	Contract	79.2	0.24	1.27	1.7	92.0	92.0
GPT-4.1	Baseline	76.7	0.28	1.43	2.5	89.3	89.3
GPT-4.1	Contract	93.3	0.15	1.13	4.2	37.5	37.5

Table 1: Stress-pilot results. FTGA, unresolved rate, Repair@1, and Repair@2 are percentages. RTT and token tax are lower-is-better; token tax is total tokens until success divided by first-attempt tokens.

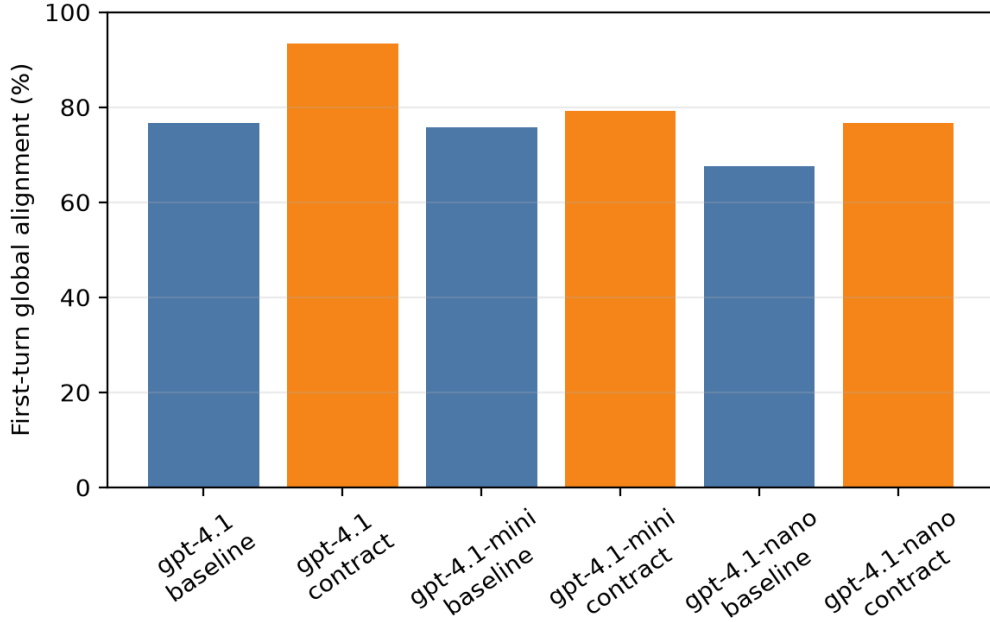


Figure 1: First-turn global alignment by model and condition on the 120-item stress pilot.

86 **Implicit editing preservation is the dominant failure mode.** Family-level analysis shows
87 aggregate baseline FTGA is 33.3% for editing preservation, with 60/90 first-turn failures
88 across models. By contrast, output-language inference is saturated at 100.0% FTGA in this
89 pilot. The contract raises editing-preservation FTGA by +63.3, +13.3, and +33.3 points for
90 gpt-4.1, gpt-4.1-mini, and gpt-4.1-nano.

91 **Cheaper models still struggle with exact preservation.** gpt-4.1-nano translates seman-
92 tically transparent quoted headings such as "Budget Review" and "Community Questions"
93 despite explicit preservation instructions. On quote-preservation items, its first-turn FTGA
94 is 70.0% in baseline and 73.3% under the contract, but the contract lowers mean RTT from
95 0.63 to 0.37 and unresolved rate from 13.3% to 3.3%.

96 **Audit.** The 72-response blinded judge audit agrees with the corrected automatic scorer on
97 71/72 sampled pass/fail labels (98.6%). The single disagreement is an editing-preservation
98 response where the rule scorer penalizes Spanish framing around English rewrites while
99 the judge accepts the answer. The audit is not a substitute for native-speaker validation, but
100 it reduces the risk that the headline trend is an artifact of the rule-based scorer.

Qualitative examples. Validated examples match the aggregate patterns: baseline gpt-4.1-nano translates English content into Spanish before repair in one editing case, and repeatedly asks for already-present Arabic-English quoted text in an unresolved quote-preservation case.

5 Related work

Recent multilingual benchmarks such as MMLU-ProX and BenchMAX broaden accuracy and instruction-following evaluation across languages (Xuan et al., 2025; Huang et al., 2025); OLA studies output-language alignment in code-switched interactions (Oh et al., 2026). Multi-turn dialogue evaluations such as MT-Bench and Chatbot Arena evaluate conversational quality (Zheng et al., 2023); our protocol instead fixes the initial contract violation and asks how many standardized repairs are needed for recovery. Re-prompt Tax is complementary: it measures the interactional cost of recovery and expands the contract beyond output language to include script, preservation, register, and locale constraints. WildChat (Zhao et al., 2024) motivates studying multi-turn repair because LLM use is diverse and interactional.

6 Limitations and ethics

This is a pilot benchmark, not a representative sample of global multilingual use. It covers only three language pairs and uses synthetic prompts. Automatic checks are reliable for exact spans and scripts, and the blinded judge audit supports the sampled language/task decisions, but native-speaker audit remains necessary before strong final claims. Token tax is a repair-burden ratio, not an API-cost estimate: the longer contract prompt increases absolute token use despite reducing repair-normalized token tax. We do not release raw user logs; all benchmark items are synthetic. The intended impact is to make hidden global usability burdens measurable, not to characterize all speakers of any included language.

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